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Investigating influential techno-economic factors for combined heat and power production using optimization and metamodeling

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HIGHLIGHTS

- Metamodel is used to study influential factors for combined heat and power production.
- Combined heat and power production is investigated with 6 techno-economic factors.
- Estimated variations in energy system cost are caused by main and interaction effects.
- Optimal factor setting maximizes electricity production and provides extra benefits.

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ABSTRACT

This paper investigates the interaction of a wide range of electricity and fuel prices and technical factors of combined heat and power production in a district heating system. A linear programming-based optimization model with the objective to minimize system cost was used to study the energy systems in the cities of Gävle and Sandviken in Sweden. The comprehensive outcomes from optimization and parametric studies have been analyzed using a polynomial-based metamodel. System costs include variable costs for the production and revenues for sale of heat and electricity. The metamodel is used as an analytical and explanatory tool to interpret input-output relationships. Municipal district heating systems of Gävle and Sandviken in Sweden are studied as an interconnected regional system with improved and new combined heat and power plants. The results show that effects from electricity and fuel prices are important, but that variations in energy system cost may also be caused by many cross-factor interactions with technical factors. A comparative system performance analysis with defined cases and optimal factor setting shows a substantial increase in the electricity production, here by up to 650 GWh annually. The profitability of investing in a new plant depends highly on the considered investment risk and electricity and fuel market prices. CO_2 emission savings by up to 466 kton annually can be accomplished if marginal electricity production from coal-condensing power plants is avoided and biofuel is released at the same time.

1. Introduction

District heating (DH) is a well-recognized concept to utilize recovered excess heat from combined heat and power (CHP) plants [1,2]. Joint production of heat and electricity in CHP plants enhances energy supply efficiency by saving primary energy and making it cheaper to provide heat and electricity than to produce them separately. A network of pipes provides the distribution of produced heat to satisfy heat demands, e.g., for space heating and hot water use. Most favorable conditions for DH appear in urban areas with many buildings, residential areas, and industrial complexes through high heat density and lower distribution costs [3]. Through effective resource use and the potential to reduce CO_2 emissions, DH systems and cogeneration of heat and electricity in CHP plants are promoted to reach short and long-term energy and emission targets [4,5].

To study energy systems, including DH and CHP, optimization models based on linear programming are widely used. Optimization models make it possible to identify optimal solutions for a combination of energy supplies by fulfilling the objective of, for instance, minimized total cost by satisfying energy demand during specified time steps. Several studies concern the optimal use of energy supplies and the integration of heat and fuel supplies in DH systems. Åberg [6] used

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typical system models to study changes in DH production by stepwise heat demand reductions. Gebremedhin [7] studied impacts of different levels of biomass prices and emission allowances on the choice of fuels and production technologies by considering new plant investments. Djuric Ilic et al. [8] investigated the cooperation between the transport and DH sectors by introducing a large-scale biofuel polygeneration production with new CHP plant installations. Amiri et al. [9] analyzed conditions for connecting CHP plants to a biogas system with the objective to present the model for biogas production systems. A number of studies used multi-integer linear programming models to study plant operations, design and operation strategy. Ommen et al. [10] considered lowering DH temperatures to study operation of power and heat production in utility plants based on network characteristics and presented by performance curves of typical CHP plant technologies. Romanchenko et al. [11] investigated the interplay between DH systems and electricity system to study optimal operating strategies for DH systems based on electricity price curves. Studies by Morvaj et al. [12] and Buoro et al. [13] used multi-objective models to reduce costs and CO_2 emissions simultaneously and investigated optimal design and operation of distributed energy systems in urban and industrial areas, respectively.

However, optimization modeling of DH systems with CHP plants depends on many factors. DH systems operate locally, and the heat production is adapted to variations in local heat demand, while a CHP production directly links the DH system to the electricity market. Uncertainties through the evolution and variation of energy prices in, for instance, a deregulated electricity market introduce difficulties in defining system conditions leading to additional approaches. Considering cogeneration system planning and by including probabilistic and simulation models, incorporating uncertainties of electricity, bio and natural gas prices with different time frames of electrical and thermal loads can be used for optimal operation of the cogeneration plant [14]. Using cases of electricity price profiles based on recent price variations and reasonable expectations on the price development enables testing of a system's sensitivity to diurnal and seasonal price variations [15]. Shadow prices (incremental cost of generating one extra unit of electricity) from a power system as input data for electricity pricing provide more exact system conditions than assumed electricity price data [7]. Considering variations in fuel and electricity prices, a statistical factorial design can be used to analyze the sensitivity of different plant capacities to evaluate the profitability and financial risk of CHP investments [16,17]. The studies also highlight the importance of considering interaction effects between techno-economic factors. Most of the mentioned energy system studies used scenarios, some sensitivity analysis or parametric studies to analyze system sensitivity and uncertainties in input data. Only a few consider interactions between economic and technical factors. With an expected increase in the complexity of operating DH systems with CHP plants, there is a need to achieve a better understanding of influential factors for CHP production.

Response surface models enable graphical solutions and are widely used to provide the understanding of input-output relationships between design factors (input variables) and response (output). Response surface models include polynomial regression models, associated with traditional response surface method and advanced Design of Experiment [18], and so-called metamodels when applied to approximate computer-based model outcomes. A detailed background to metamodeling and response surface method with applications can be found in [19]. The approach used in this study is illustrated in Fig. 1. To the best of the author's knowledge, this is the first time the approach of energy system (including DH and CHP), optimization model and metamodel is used. Principally, an energy system with supply and delivery is described by an optimization model. Specific model input data in the form of design factor level combinations ($x_1, \dots, x_k$), also referred to as factor setting, are used to generate a limited set of model runs (scenarios) and output data (y), which in turn are used to fit a metamodel

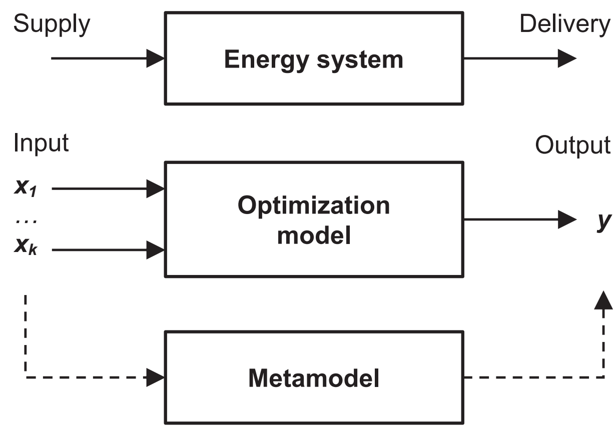


Fig. 1. Illustrated method approach with energy system (including district heating and CHP), optimization model, and metamodel.

that approximates the model's input-output transformation.

The aim of the study was to investigate influential factors for CHP production by using an optimization tool and metamodeling. The linear program MODEST was used for optimizations. Metamodeling includes a polynomial-based metamodel to approximate cost-minimized model outputs, and Box-Behnken experimental design to generate the limited set of scenarios. Municipal DH systems were connected into a larger regional system and studied with improved and new CHP plant installations. A comparative system performance analysis with defined cases based on optimal factor setting was used to show impacts on heat and electricity production, industrial heat use, energy system cost, and CO_2 emissions. CO_2 emissions were accounted for using a Nordic electricity system perspective with models for marginal electricity, limited biofuel, and electricity production mix. Different discount rates were used to evaluate the investment opportunity of improved and new plants.

The paper is organized as follows. Section 2 includes a background of studied cases with local and regional DH systems and description of energy supplies. Section 3 describes the linear program MODEST and the modeling of the studied energy system with input data. Metamodeling also includes generated scenarios and design factors. Differences to separate systems and models are highlighted as well as model validation, CO_2 emission accounting and calculation of investment opportunity. Section 4 shows the metamodel for energy system cost, graphical solutions and changes in system performance. Section 5 includes further considerations to develop the study and findings and conclusions are presented in Section 6.

2. Energy system description and defined cases

The studied energy systems deliver heat for DH in the cities of Gävle and Sandviken with a combined population of about 130,000 inhabitants. The systems are located close to each other in Gävleborg County in east-central Sweden (Fig. 2). Both systems are owned and operated by a municipal energy company, Gävle Energi AB (GEAB) and Sandviken Energi AB (SEAB) respectively. Each system currently uses heat supplies from its own biofuel CHP plant, mainly from November to March. Heat-only plants and boilers (HOBs) are used during summer (Sandviken) and for increased DH demand. Industrial heat is used in Gävle for base heat loads throughout the year. Heat supplies come from a pulp and paperboard mill, mainly as excess and evaporator heat, and a jointly owned CHP plant (GEAB, mill). The biofuel-fired industrial CHP plant produces steam and electricity for the mill, while heat from the flue-gas condensation (FGC) is supplied for DH, but also from a (steam) condenser to cover peak heat loads. A fossil fuel-based HOB is also used for peak demand from both pulp and paperboard mill and DH. Industrial stand-by electric boilers are not considered here. The energy

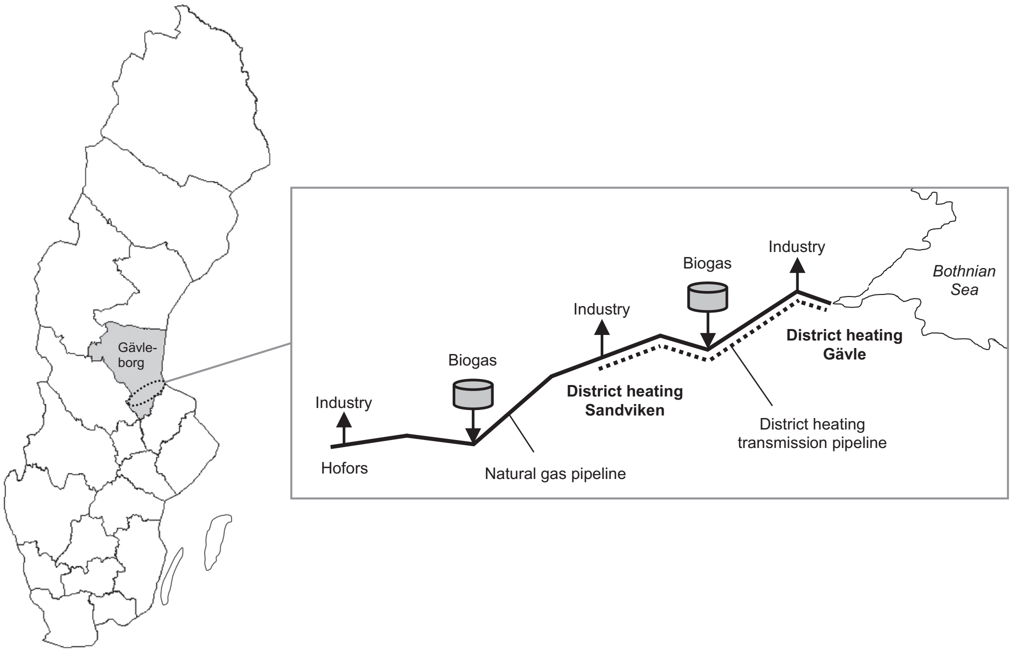


Fig. 2. Sweden and geographical location of Gävleborg County with illustration of outlined regional gas network and district heating transmission pipeline to connect municipal energy systems in Gävle and Sandviken (distance about 20 km). The gas supply is planned via Gävle by LNG (liquefied natural gas) terminal or floating storage facility and delivered via pipeline to industries. Distribution of locally produced biogas is also considered for the future.

companies are presently investigating a connection of the municipal energy systems into a larger regional system (Fig. 2). The reasons are, in part, the increased utilization of industrial heat and the effective use of plant heat production [20]. In addition, there is also interest in establishing a regional gas network to supply industries with natural gas (Fig. 2). A driving force is the reduction of environmental impacts through conversion of fuel oil and LPG (liquefied petroleum gas) to biogas. Industries and municipal energy systems were both already subjects of previous studies analyzing economic potential of locally deregulated and regional heat markets with extended cooperation between industries and energy companies [21–23].

2.1. Studied regional case with potential energy supply (PES)

The existing municipal energy systems were here assumed as connected into a larger regional system and studied with new potential energy supplies of heat and electricity, further denoted as PES. The regional case with PES was used to investigate opportunities for increased electricity production and includes a new natural gas combined-cycle (NGCC) CHP plant with technical specifications of low to high range for electric power sizes and alpha values. Electric power sizes may indicate a proper plant output capacity, while applied alpha values (ratio between electricity and heat output) may show a preferable operation strategy, that is, more heat or electricity production. A natural gas supply was assumed via the outlined network that matches each chosen electric power size. Moreover, the PES also includes an alpha value range for existing biofuel CHP plants with current and increased values to improve electricity output capacities. Together with technical specifications, relevant economic data with price ranges for supplied biofuel and natural gas and sold electricity were used to reflect changing system conditions for the operation of the CHP plants. Technical specifications and system conditions were performed by numerous scenarios based on the experimental design.

2.2. Reference and base case

The reference and base case were used to analyze changes in the heat supply for DH and CHP electricity production and to assess economic and environmental impacts through system interconnection and PES. The reference case represents today's energy systems in Gävle and Sandviken. The separate systems were also used to validate energy system models, because the outlined regional system does not yet exist. The base case considers today's energy systems as already interconnected but with existing energy supply.

3. Methods and input data

3.1. MODEST optimization model

MODEST (Model for Dynamic Energy System with Time-dependent components and boundary conditions) is a deterministic energy system optimization model based on linear programming. Linear programming is used to determine the optimal system design and operation that satisfies energy demand at lowest system cost. Optimal design and operation are obtained by the best combination of system components and energy flows adapted to the specified properties and conditions. Relationships between system components and energy flows are described by linear relations. A bottom-up modeling approach enables almost arbitrary combinations of linear relations for supply, conversion, distribution, and demand of energy. The modeling is supported by a user-friendly interface for input and output data. Input data depends on required level of detail and consists of time division, supply and demand, and different types of plants and equipment. Output data contains files for documentation and the optimization software that performs the cost minimization. To generate the output by a model run usually takes seconds on an ordinary computer workstation. The optimized output is precisely determined without any random variation, that is, a model run with given input will always produce the same output through known linear relations and input data without random variation. Results include graphs and spreadsheets with optimal plant use, costs and revenues, emissions, and all modeled energy flows during specified time periods.

MODEST was developed at the University of Linköping [24,25] and applied on many previous studies to optimize energy systems for DH with CHP operation and industrial heat supplies and to evaluate economic and environmental consequences of energy system changes at local and regional levels [22,24–29]. The 2016 version of MODEST and optimization software Cplex 12.4.0.0 were used in the study.

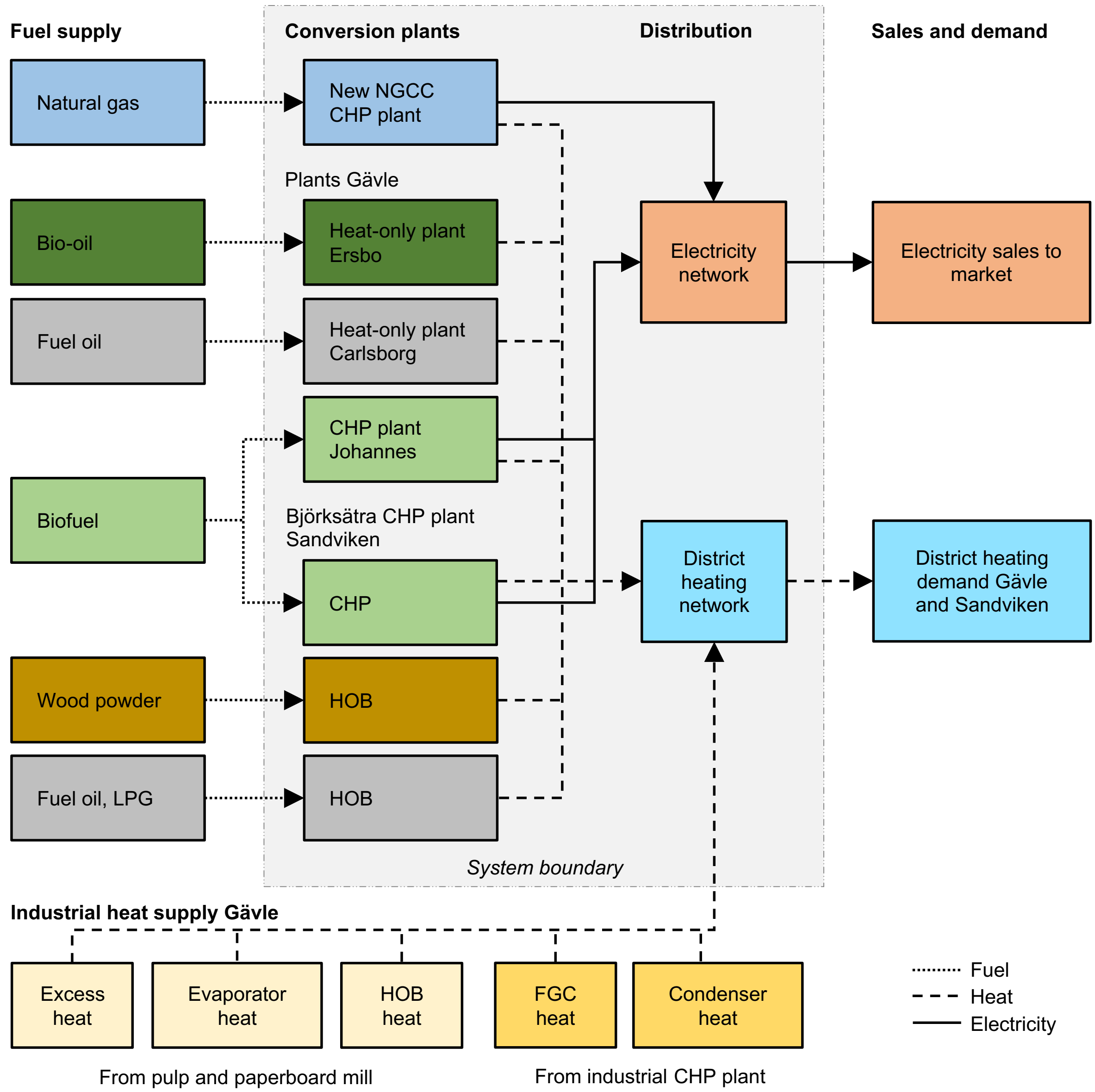


Fig. 3. Schematic illustration of the MODEST model structure and considered system boundary for the interconnected regional energy system Gävle and Sandviken with existing and new (NGCC CHP) plant installations.

3.2. Energy system model for regional case with PES

The regional energy system with PES was modeled as an integrated system with expanded geographic boundaries including existing and new plant installations as well as distribution networks (Fig. 3). A system integration approach provides a joint production optimization and avoids sub-optimizations, that is, municipal energy systems and new NGCC CHP plant are connected but optimized separately. Different network conditions and transmission capacity limits were ignored here and heat supplies for regional DH were assumed as equally balanced without restrictions in the model. Average distribution losses of 12% were assumed for heat deliveries to Gävle and Sandviken via the regional DH network.

As presented in Fig. 3, the model structure used in MODEST describes the system by a set of components (nodes). Nodes represent supply of fuel and industrial heat, conversion plants with heat and electricity production, distribution of energy via networks, sale of electricity and required DH demand. Nodes for plants or units of a plant (Sandviken) are solely described by an energy balance of its input and output. The DH network node represents connected networks and considers the distribution losses. Each node relates to one or more energy flows (fuel, heat, electricity). Energy flows from supply nodes to conversion and network nodes and further to market and demand nodes. Produced electricity is delivered without losses to the Nordic/Baltic electricity market via the national electricity grid. The time-dependent DH demand that must be satisfied is represented by merged system heat loads for Gävle and Sandviken.

3.2.1. Time division and district heating demand

The use of an accurate time division is important for the optimization of heat supply sources and in particular for the cogeneration of electricity in a CHP plant, because the amount of electricity that can be cogenerated depends on the heat that is needed to satisfy DH demands. A time division should therefore reflect seasonal and daily heat load variations that must be met by the heat supply sources. Seasonal heat load variations are significant between winter and summer in a country like Sweden with a colder climate. Daily heat load variations may describe normal human behaviors (e.g. space heating, hot water use) during daytime and nighttime. However, a typical annual DH demand is here considered to be 952 GWh/a and dominated by the demand in

Gävle. The annual time division used contains 88 time steps and together with heat loads is shown in Table A1. Time steps include seasons and diurnal periods. Seasons with increased space heating demand (November to March) include twelve diurnal periods to describe realistic hourly variations on weekdays, weekends and peak days. Peak days reflect top heat loads and are important to consider maximum heat supply capacity needed in the system. Seasons with low DH demands (April to October) include only four diurnal periods.

3.2.2. Fuel and industrial heat supply

The supply of fuels to heat-only and CHP plants was modeled without capacity restrictions. A natural gas supply is given by the outlined network. Industrial heat supply from Gävle was assumed as available throughout the year but may vary, for instance, through maintenance periods in the industry. The various heat flows to the DH network were restricted by maximum net capacities. The fossil fuel-based heat production was assumed as HOB heat supply in the model.

3.2.3. Energy balance at conversion plants

A modeled energy balance at a conversion plant or unit of a plant is the sum of the node outflow(s) and equal inflow multiplied by the efficiency. Efficiencies are either total plant or boiler (Sandviken) performance. Power capacities specify maximum node outflow(s) and alpha values describe outflow relationships. Each energy balance implies the assumption of constant power capacities without considering minimum heat load requirements. Efficiencies and chosen alpha values are also fixed and supposed to be independent of a plant's or unit's energy output.

In Fig. 3, heat-only plant or HOB nodes are balanced by a simple fuel inflow and heat outflow relationship. CHP plant nodes (new plant, Gävle) include fuel inflow and simultaneous electricity and heat outflows, and with an assumed control surface around such a node, an energy balance can be expressed as

$$P_f = P_e + P_h + P_l \tag{1}$$

where P_f is the supplied fuel in the form of generated heat by combustion processes (e.g. biofuel boiler or gas turbine), P_e and P_h are the power capacities for electricity and heat output, and P_l is the energy loss which by knowing the efficiency η can be defined as

$$P_l = (1 - \eta)\cdot P_f \tag{2}$$

The relationship of the electricity and heat output capacity is further given by the alpha value (α).

$$\alpha = \frac{P_e}{P_h} \tag{3}$$

The CHP node (Sandviken) includes power capacities for cogeneration and heat-only production. Heat generated (P_f) is balanced with outflows from a cogeneration and heat-only node (not shown in Fig. 3). The CHP node's output relationship given by α is here related to the cogeneration node. Alpha values for biofuel CHP plants are gross values without considering FGC heat output. Recovered heat from biofuel combustion is here defined as heat supply with a capacity related to P_f to reduce a plant's biofuel use (not shown in Fig. 3). The biofuel use reduction is described by a time-independent factor of −0.11 for the CHP node in Sandviken and −0.2 for the CHP plant node in Gävle.

3.2.4. Determining heat and electricity output with different alpha values

Different alpha values were used to adjust a CHP plant's electricity and heat output capacity. Based on an initial state of a plant's technical specification, the fuel energy supply was calculated and further restricted when changing alpha values. Initial states contain current power capacities and alpha values for existing biofuel CHP plants, and electric power sizes combined with highest alpha value for new NGCC CHP plant (see Table 1). According to energy balance and by combining Eqs. (2) and (3), the power capacity P_e of a CHP plant with changed alpha value was then determined by

$$P_e = \frac{\eta\cdot P_f}{\left(1 + \frac{1}{\alpha}\right)} \tag{4}$$

where P_f is here the maximal fuel energy supply capacity based on initial state. For instance, if the initial state of the NGCC CHP plant with P_e of 40 MW, α of 1.2, η of 90%, and calculated maximal P_f of 81 MW changes to an α of 0.8, the P_e for the electricity output decreases to 32.6 MW. The power capacity for the heat output (P_h) is given by Eq. (3).

3.2.5. Calculating energy system cost

The energy system cost was calculated as the sum of annual variable costs and revenues. No investment costs, fixed costs or costs related to a plant's output power were considered (see also Section 3.6, calculation of investment opportunity). Variable costs (C) were accounted to produce electricity and heat (CHP, HOB) and to supply industrial heat (IH). The production cost of a plant or unit of a plant includes fuel purchase, operation and maintenance (O&M). Each industrial heat supply includes purchase price, and costs for pumping and maintenance. Revenues (R) were calculated as negative variable costs without income tax and consider sales of electricity to the market and heat for DH. The sum of annual costs was determined as present value and optimized for a period of ten years with a common discount rate of 6% [32]. The discount rate is a real value to avoid influences from inflation and was supposed to be constant during the optimization period.

The sum of annual costs and revenues (C_a) is expressed as

$$C_a = \sum C_{CHP} + \sum C_{HOB} + \sum C_{IH} - R_{Heat} - R_{Electricity} \tag{5}$$

and a general expression for calculating variable costs (or revenues R) is

$$C = E\cdot c\cdot \tau \tag{6}$$

where E is the size of an energy flow (MW), c is the specific cost (EUR/MWh) for production or industrial heat supply, and τ represents the number of obtained hours. Production costs are attached on fuel inflows or electricity outflow (NGCC CHP). Revenues are gained by aggregated energy flows delivered via networks. For instance, the heat flow from the DH network is attached with a sale price of 51 EUR/MWh and the size of the heat flow (excl. losses) is given by the time-dependent system heat load. The electricity flow size depends on a cost-effective cogeneration and is only restricted by the CHP plants' electric power capacities.

3.2.6. Techno-economic input data and limitations through low-priced industrial heat supply

In Tables 1 and 2, input data for Gävle are based on information from GEAB (personal communication), while data for Sandviken is related to a previous study [31] but with modified heat capacities according to SEAB [33]. Observed techno-economic data for improved and new CHP plants with specified variable value ranges are based on common literature. Remaining data for fuel and industrial heat supply, conversion plant installations, and heat sale price are hold constant single variable values. Economic data is converted by 9.1 SEK = 1 Euro (EUR) and excludes VAT.

Apart from current biofuel CHP plant specifications, each technical value range was evaluated from an initial data set [32] and further adjusted by model trial runs. According to NGCC CHP plant, lower and upper electric power sizes were governed by the annual heat production. Selected sizes cover the presumptions that the plant should produce less heat than biofuel CHP plants, but also has the potential to substitute the entire industrial heat supply. The upper level at 90 MW avoids an oversizing of the plant. Sizes were also evaluated by different alpha values. The final alpha value range was chosen with the intention to indicate changes in the operation strategy. Higher alpha values may usually be considered for NGCC CHP plants [32]. In contrast, the

Table 1
Techno-economic input data for existing and new plant installations and industrial heat supply.

Origin, type	Heat (electric) power (MW)	Alpha value (–)	Efficiency (%)	Fuel and heat supply	Production and heat supply cost (EUR/MWh)
New CHP plant					
NGCC CHP	(40–90)[a]	0.8–1.2	90	Natural gas	19.7–23.7[b,c]
Plants Gävle					
CHP plant Johannes	75[d] (21)[e]	0.28–0.52	88	Biofuel	20.3–24.3[f,h,g]
Heat-only plant Ersbo	76	–	85	Bio-oil	132
Heat-only plant Carlsborg	60	–	85	Fuel oil	109[i]
Björksätra CHP plant Sandviken					
CHP	40[j] (5.2)[k]	0.28–0.52	85	Biofuel	20.3–24.3[f,h,g]
HOB	20	–	85	Wood powder	33[l]
HOB	50	–	90	LPG	54
HOB	25	–	90	Fuel oil	109[i]
Industrial heat supply Gävle[m]					
FGC heat	20	–	100	Industrial CHP plant	2.2
Condenser heat	8	–	100	Industrial CHP plant	18.7
Excess heat	23	–	100	Pulp and paperboard mill	1.6
Evaporator heat	36	–	100	Pulp and paperboard mill	8.2
HOB heat	120	–	100	Pulp and paperboard mill	99

[a] Electric power sizes are associated with alpha value 1.2.
[b] Consider historical price variations from 2010 to 2015 based on European spot market (UK NBP) [35].
[c] Includes 2.7 EUR/MWh O&M cost, but is allocated for each MWh of produced electricity and based on [32].
[d] Additional 22 MW heat power comes from integrated FGC.
[e] Current heat and electric power size associated with alpha value 0.28.
[f] Biofuel prices consist of fractions from wood chips (50%), by-products (15%), recycled (20%), sod peat (15%).
[g] Lower value is based on historical data from 2010 to 2015 [36] and upper value is an estimated future average price based on [37].
[h] Includes 2.3 EUR/MWh O&M cost (incl. FGC) allocated for each MWh of used biofuel and based on [32].
[i] Adapted average price for residual fuel oil with taxes [35] and O&M cost [38].
[j] Consists of two biofuel boilers each with 20 MW and additional 8 MW comes from integrated FGC.
[k] Existing net heat supply capacity from boilers for cogeneration is 18.6 MW with current alpha value 0.28.
[l] Upgraded price for densified wood fuels [36] with O&M cost based on [38].
[m] Heat power is considered as net supply capacities without losses and efficiency 100%.

Table 2
Energy sale prices.

Type	EUR/MWh
Heat	51[a]
Electricity	36–44[bc]

[a] Average price for Gävle [39] and Sandviken [40] from the year 2017.
[b] Historical price variations from 2010 to 2015 are based on Nord Pool Spot market [41].
[c] Prices are only market values without distribution charge and taxes.

additional alpha value for the biofuel CHP plants reflect higher levels from common Swedish examples. Differences in existing plant specifications such as output capacity were not considered here. Efficiencies were assumed as average values and supposed to be constant, irrespective of chosen electric power size and alpha value.

The evaluated economic value ranges imply the following assumptions: variable O&M costs for CHP production are constant; purchase prices of biofuel and natural gas and sale prices of electricity are annual average values and indicate changing system conditions. Historical price variations were limited by model trial runs to reduce uncertainties in new plant utilization. Uncertainties were given by the presence of low-priced industrial heat supply. Moderate natural gas prices and sufficient high market values for electricity were necessary to make a NGCC CHP plant operation profitable. A profitable plant operation was needed to secure balanced effects when changing inputs. Balanced effects are required to produce adequate metamodel approximations. Hence, a center-based price level was further expanded by a 10% increase and decrease to determine lower and upper limits of price ranges. Energy and emission costs for natural gas use (e.g. through energy tax, CO_2 tax, emission rights [32]) were not considered; instead a higher fuel price may reflect such extra costs. Biofuels are assumed as the same for both CHP plants and prices are compounded prices from biofuels commonly used in Gävle and Sandviken (Table 1). Small biofuel price variations were expanded by a higher price level, which may be motivated by increased future competition on biofuel markets [34]. Electricity certificates[1] were not included in electricity sales from aging biofuel CHP plants, nor for considered improvements to increase their power capacity.

3.2.7. Differences to reference and base case models

Reference and base case models are similarly structured in MODEST as illustrated in Fig. 3. According to the reference case, each system is represented by its own energy system model with respective fuel and heat supply, conversion plants, and DH demand according to Tables 1 and A1. Average distribution losses here are local losses for heat deliveries in Gävle and Sandviken. The model for the base case corresponds to the energy system model with PES, but without new plant and associated energy flows.

3.3. Energy system model validation with respect to real data

The validation was performed with separate energy system models of the reference case. A validity of reference case models may be considered as representative for the validity of the base case model with similar model structure and input data, but merged heat loads. A use of separate model outcomes provides a comparison with real data. Real data represent average values and were considered for different years depending on available data. A use of average values is motivated by common annual variations in DH demands and their effect on increased

[1] Swedish subsidies to promote renewable electricity production with a common time period of 15 years [59].

Table 3
Energy system model outcomes in comparison to real data for Gävle and Sandviken (GWh/a).

Origin	Heat production	Industrial heat supply	Total production and supply of heat	Delivered heat	Cogenerated electricity
Gävle data[a]	252	489	741	662	61
Model	259	561[b]	820	722	56
Sandviken data[c]	254	–	254	223	18
Model	262[d]	–	262	231	24

[a] Average values from the years 2014–2016 based on GEAB [42].
[b] Mainly excess heat (201 GWh/a), FGC heat (151 GWh/a), and evaporator heat (204 GWh/a).
[c] Statistical average values for the years 2012–2014 based on [43,44].
[d] Includes about 10 GWh/a from wood powder HOB and 3 GWh/a from gas HOB.

or decreased heat production and industrial heat use. A relative deviation was here measured for adequate model outcomes. The relative deviation is the difference in respective annual contributions of heat production, industrial heat supply, and cogenerated electricity. Prices for electricity and heat sales in the models were 36 and 51 EUR/MWh, respectively, and a production cost of 20.3 EUR/MWh was used for biofuel CHP plants.

In Table 3, the heat production of the models is dominated by CHP heat outcomes. Only the Sandviken model includes HOB heat production. Compared with real production data, small amounts of heat from heat-only plants occur in Gävle, while a wood powder HOB heat production usually has a higher portion in Sandviken. A HOB heat production in the model increases remarkably if CHP production is shut down during summer in Sandviken (cf. Ref. [31]). Also, the amount of cogenerated electricity decreases. However, Gävle model outcomes for heat production and industrial heat supply show a relative deviation of ± 2% in accordance with total production and supply of heat. Only a small relative deviation occurs between total production/supply and delivered heat. The amount of cogenerated electricity in relation to heat production, including heat-only production, deviates by 2% for both energy system models.

3.4. Metamodeling and scenarios for PES

A basic approach for MODEST is to build a metamodel of the input-output transformation that is formulated by the linear relations of the system components. The modeled energy system with PES is treated as a black box and the metamodel is then used as an analytical and explanatory tool to interpret the relationship between inputs and output. An overall form to express the relationship between a single output measure y and input variables x of a computer code with function f is

$$y = f(x_1, x_2, \cdots, x_k) \tag{7}$$

The mathematical function of the computer code may be fully understood by the bottom-up modeling approach as in MODEST. On the other hand, with the arrangement of many input variables and their value levels, the analysis of each possible variable combination is time-consuming and requires many model runs. At the same time, it may provide little insight into the relationship and interactions between the input variables. In contrast, the analytical form of a metamodel with function g providing such insight can be expressed by

$$Y = g(x_1, x_2, \cdots, x_k; b_1, b_2, \cdots, b_k) \tag{8}$$

and gives

$$Y = y + \epsilon \tag{9}$$

where Y is the approximated response and b are estimated parameters representing expected changes in the response per unit change of an input variable, presuming that remaining variables are held constant. The number of parameters depends on the number of input variables and the form of the metamodel that approximates the function, for instance, a second-order polynomial in $x_1, x_2, \ldots, x_k$. Differences between the metamodel's analytical form and computer code output can be expected and considered by the approximation error ε for deterministic models. Uncertainties in approximations are considered by statistical measures through the metamodel's validity testing.

Metamodeling can be classified according to applied statistical techniques in experimental design for generating the input-output data, the model used to represent the data, and the fitting of the model to the observed data [45]. Kleijnen and Sargent proposed a methodology with ten steps for metamodeling [46]. Some of the steps are considered here to develop the metamodel, and a summary of the statistical techniques and generated scenarios is shown in Fig. 4. Response variable Y is the optimized MODEST output, i.e., minimized energy system cost. Design factor levels low and high correspond to the techno-economic variable value ranges of improved and new CHP plants and are price of biofuel (x_{pb}), natural gas (x_{pn}), electricity (x_{pe}), electric power size (x_{en}), and alpha values (x_{an}) and (x_{ab}). The latter consider biofuel CHP plants and middle values are average numbers. Statistical software (here: Minitab version 2017) is commonly used for a metamodeling process.

According to the second-order polynomial regression metamodel in Fig. 4, X_i and X_j are the original factors expressed as coded variables, β_o is the model intercept coefficient and β_i ($i = 1, 2, 3, \ldots, k$), β_{ij} ($i = 1, 2, 3, \ldots, k$; $j = 1, 2, 3, \ldots, k$) are regression coefficients for the linear, quadratic and two-factor interaction terms. Linear terms indicate the importance of the factor's main effect on the response by their level change. Two-factor interaction terms show how the level change of a factor and its effect on the response depends on another factor's level.

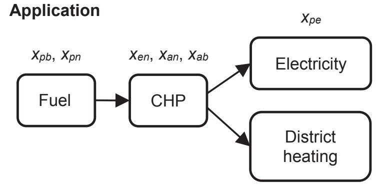
The Box-Behnken response surface design is a classic experimental design to estimate parameters in a second-order polynomial model [47]. The three-level design consists of a subset (fraction) of factorial combinations from a 3^k factorial design. With a factor arrangement of $k = 6$, the fractional design requires 49 model runs (including one center point) to generate experimental input-output data, instead of 729 in a (full) factorial design. Each factor has evenly spaced levels with settings low, middle, and high. Factor levels are placed on a common scale to make effects of the factors comparable. For instance, specified fuel price levels (EUR/MWh) or electric power levels (MW) are standardized to coded variables (-1, 0, 1). The coding formula in Fig. 4 standardizes factor levels into coded variables (X_i), where x_i is the original factor level, and x_{iM} is the average of the low (x_{iL}) and high (x_{iH}) factor levels ($i = 1, 2, \ldots, k$). The design region defined by the coded variables is spherical in shape where all factor level combinations are geometrically on the center of the edges of a cube with same distance to the design center. Level settings where all factors simultaneously have their lowest or highest level are therefore not considered. Replications, randomization or blocking in the experimental design are irrelevant when using a deterministic model [48].

3.5. CO_2 emissions accounting

CO_2 emissions were accounted for direct and indirect emissions by using emission factors (Table 4). Direct emissions were considered for the fuels used in the conversion plants to produce heat and electricity and for the use of industrial heat. Indirect emissions were used to reflect impacts of local cogenerated electricity and biofuels used on other

Design factor		Low (-1)	Middle (0)	High (1)
x_{pb}	EUR/MWh	18	20	22
x_{pn}	EUR/MWh	17	19	21
x_{pe}	EUR/MWh	36	40	44
x_{en}	MW	40	65	90
x_{an}	-	0.8	1	1.2
x_{ab}	-	0.28	0.4	0.52

Application



Coding formula

$$X_i = 2 \cdot \frac{x_i + x_{iM}}{x_{iH} - x_{iL}}$$

Polynomial regression metamodel

$$Y = \beta_0 + \sum_{i=1}^{k} \beta_i X_i + \sum_{i=1}^{k} \beta_{ii} X_i^2 + \sum_{i \neq ij}^{k} \beta_{ij} X_i X_j$$

Run	X_{pb}	X_{pn}	X_{pe}	X_{en}	X_{an}	X_{ab}	Run	X_{pb}	X_{pn}	X_{pe}	X_{en}	X_{an}	X_{ab}
1	-1	-1	0	-1	0	0	**26**	1	0	0	-1	-1	0
2	1	-1	0	-1	0	0	**27**	-1	0	0	1	-1	0
3	-1	1	0	-1	0	0	**28**	1	0	0	1	-1	0
4	1	1	0	-1	0	0	**29**	-1	0	0	-1	1	0
5	-1	-1	0	1	0	0	**30**	1	0	0	-1	1	0
6	1	-1	0	1	0	0	**31**	-1	0	0	1	1	0
7	-1	1	0	1	0	0	**32**	1	0	0	1	1	0
8	1	1	0	1	0	0	**33**	0	-1	0	0	-1	-1
9	0	-1	-1	0	-1	0	**34**	0	1	0	0	-1	-1
10	0	1	-1	0	-1	0	**35**	0	-1	0	0	1	-1
11	0	-1	1	0	-1	0	**36**	0	1	0	0	1	-1
12	0	1	1	0	-1	0	**37**	0	-1	0	0	-1	1
13	0	-1	-1	0	1	0	**38**	0	1	0	0	-1	1
14	0	1	-1	0	1	0	**39**	0	-1	0	0	1	1
15	0	-1	1	0	1	0	**40**	0	1	0	0	1	1
16	0	1	1	0	1	0	**41**	-1	0	-1	0	0	-1
17	0	0	-1	-1	0	-1	**42**	1	0	-1	0	0	-1
18	0	0	1	-1	0	-1	**43**	-1	0	1	0	0	-1
19	0	0	-1	1	0	-1	**44**	1	0	1	0	0	-1
20	0	0	1	1	0	-1	**45**	-1	0	-1	0	0	1
21	0	0	-1	-1	0	1	**46**	1	0	-1	0	0	1
22	0	0	1	-1	0	1	**47**	-1	0	1	0	0	1
23	0	0	-1	1	0	1	**48**	1	0	1	0	0	1
24	0	0	1	1	0	1	**49**	0	0	0	0	0	0
25	-1	0	0	-1	-1	0							

Fig. 4. Summary table of statistical techniques with design factors for improved and new CHP plants (flow chart) and generated scenarios (run) by the Box-Behnken experimental design.

emission sources in an overall electricity system. Different accounting methods correspond to models for assessing CO_2 emissions, that is, marginal electricity combined with limited biofuel according to [49] and electricity production mix. Calculations were idealized by excluding energy use and associated CO_2 emissions from the fuel supply chain (production, storage, transportation), operation of plants and networks, and losses from industrial heat supply and delivery of electricity to the market.

3.5.1. Direct emissions from fuel and industrial heat use

Direct emissions from fuel use correspond entirely to the fossil fuel combustion. The use of biofuel was considered CO_2 neutral, since a contributed amount of CO_2 through biofuel combustion can be expected to be equivalent to the absorbed amount of CO_2 from the atmosphere over the life-cycle of the plant's growth [53]. No CO_2 emissions were accounted for the use of industrial heat, since all emissions can be assumed as allocated to a primary source (e.g. industrial process) located outside the system boundary.

3.5.2. Marginal electricity model

The model considers the marginal source of electricity responding to changes in the supply and demand in a deregulated electricity market. Since Sweden, like other Nordic and Baltic countries, is part of a deregulated electricity market (Nord Pool), the accounting for CO_2 emissions from locally produced electricity referred to a national level is less relevant. A spot market such as Nord Pool balances the supply and demand of electricity and calculates electricity prices of the coming day. The concept is that electricity is priced differently through the moment of consumption and associated short-run costs [54]. Short-run costs reflect additional costs for a specific supply technology to produce one unit of electricity. Considering the installed supply technologies in the Nordic/Baltic electricity market with ascending short-run costs, a price formation as illustrated in Fig. 5 shows the merit order of a typical annual electricity supply. Hydro and nuclear power with low short-run costs dominate the electricity supply. Fossil fuel-based condensing power has generally higher short-run costs and may be used when electricity consumption increases. Oil-condensing and gas turbines with the highest short-run costs may only cover peak demand. With an electricity consumption of about 410 TWh/a in 2016 [55], the marginal production responding to changes in the supply and demand will be coal-condensing power (CCP) plants. A new installed CHP plant for DH may avoid CO_2 emissions per MWh electricity from such CCP plants on a short-term perspective (years). On a long-term perspective (decades), including ambitious energy and emission targets in the European Union

Table 4
CO_2 emission factors[a] for direct and indirect accounted emissions (kg/MWh)[b].

Accounting method	Direct					Indirect	
	Fuel oil[c]	LPG	Biofuel[d,e]	Natural gas	Industrial heat	Biofuel	Electricity
Marginal CCP, biofuel limited	279	234	0	202	0	341	930
Marginal NGCC, biofuel limited	279	234	0	202	0	202	400
Nordic electricity production mix	279	234	0	202	0	0	100

[a] Factors are based on [49,50] for fuels and [51,52] for electricity.
[b] Conversion factor between kg/MWh and kg/MJ is 3600.
[c] Consider residual fuel oil.
[d] Include wood powder and bio-oil.
[e] Exclude emissions from biofuel production systems.

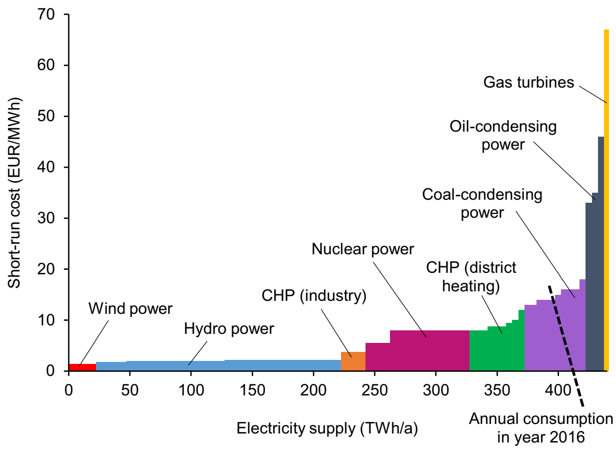


Fig. 5. Schematic price formation at the Nordic/Baltic electricity market with short-run costs, annual electricity supply from respective technology, and outlined annual electricity consumption (adapted from [51,52]).

[5], investments in more efficient and lean carbon technologies may occur and replace current CCP plants as marginal source. The alternative marginal source considered can be assumed as NGCC-condensing power (NGCC) plants [56].

3.5.3. Limited biofuel model

By assuming biofuel as a limited resource, the model can be used to evaluate effects of changes in local biofuel use on applications that use, or may use, biofuels to replace fossil fuels in power plants, for instance, through co-firing [57,58]. A resource limitation may be hypothetical, because restrictions in use of a biofuel may primarily depend on market prices, available supply infrastructure, and the type of biofuel treated on the market. However, with the assumption of cost-effective and balanced biofuel trade markets for supply and demand in Europe, a decrease of locally used biofuels provides the opportunity to replace fossil fuel-based power production somewhere else in the European electricity market. Release of a unit of biofuel in the studied systems is therefore debited by the same amount of CO_2 emissions as the replaced fossil fuel per MWh fuel. In contrast, the increased use of biofuels may reduce the opportunity to replace fossil fuels and consequently increase CO_2 emissions.

3.5.4. Electricity production mix model

An electricity production mix model considers a historical mix of supply technologies on a national or international account. The historical production mix is basically used for an average emission accounting [51]. Average emissions are here considered for a Nordic electricity production mix. Due to average emission accounting, the model does not reflect consequences of changes in a Nordic electricity system, neither in the short nor long term, but it can be used as a complement to marginal models to estimate CO_2 emission effects on the overall electricity system.

3.6. Calculation of investment opportunity

Investment costs for improved and new CHP plants were not included in system cost calculations. Instead an investment opportunity (IO) was calculated based on energy system cost. It can be used to estimate how high investment costs may be to make such an investment profitable for the energy companies in Gävle and Sandviken. The opportunity to invest in improved and new CHP plants was given by the difference in energy system cost, according to:

$$IO = C_{a(reference;base)} - C_{a(regional\quad PES)} \tag{10}$$

where $C_{a(reference;\ base)}$ denotes the sum of annual costs and revenues for reference case or base case, and $C_{a(regional\ PES)}$ is the sum of annual costs and revenues for the regional system with PES. The use of both reference and base case makes it possible to separate the system integration effect.

For each case, the energy system cost was calculated with additional discount rates of 2% and 10% (see also Section 3.2.5). The use of various discount rates makes different risks (e.g. capital-intensive technology) and uncertainties (e.g. changing system conditions) associated with a potential investment comparable. A higher discount rate implies higher investment risks and determines lower present values, and hence reduces incentives for investments.

4. Results and discussion

4.1. Metamodel for energy system cost

After performing the scenarios by MODEST, the fitting data set with 49 minimized energy system costs was used to produce a full quadratic model. A full quadratic model includes all estimated effects of a factor besides the interceptor. Factor effects were further ranked according to relative importance. The relative importance can be expressed by a coefficient's absolute value through standardized factor inputs. Absolute values for main and interaction effects remain the same during the fitting. Each absolute value represents a mean change in the response. Initially, the factor effect with highest magnitude was chosen first. Each further added factor effect was then selected according to its magnitude in descending order. The selection was performed as long as adjusted coefficient of determination (adjusted R^2) was improved. The validity measure Absolute Relative Error (ARE) was computed to support selections. The fitted metamodel in Eq. (11) has the best result for adjusted R^2 at 99.88%; corresponding R^2 is 99.94%. Main effects have the largest impact on the adjusted R^2 (95.62%), followed by two-factor interactions (3.86%) and squared effects. Some selected factor effects with a relatively low absolute value also have rather small impact on adjusted R^2. As a basis for comparison, the analysis of variance for the metamodel is shown in Table A3.

The metamodel is complex showing that estimated main effects of all design factors together with many two-factor interactions affect the response Y. All main effects have the signs expected and are in order of importance β_{pe}, β_{pn}, β_{en}, β_{an}, β_{pb}, and β_{ab}. The most important main effects are related to economic factors (X_{pe}, X_{pn}) followed by technical factors (X_{en}, X_{an}). The sign of the main effects indicates that lowest energy system cost is reached by an optimal factor setting of low fuel prices (X_{pb}, $X_{pn} = -1$), high electricity price ($X_{pe} = 1$), maximum electric power size ($X_{en} = 1$) and high alpha values (X_{an}, $X_{ab} = 1$). Most of the results would be expected.

$$\begin{aligned} Y = &-41.6764 + 0.4730X_{pb} + 1.6259X_{pn} - 1.6811X_{pe} - 1.0950X_{en} - 0.9332X_{an} \\ &- 0.2832X_{ab} - 0.1598X_{pn}^2 - 0.0746X_{pe}^2 + 0.1046X_{en}^2 + 0.0272X_{an}^2 \\ &- 0.0275X_{ab}^2 - 0.2121X_{pb}X_{en} + 0.0526X_{pb}X_{an} + 0.0316X_{pb}X_{ab} \\ &+ 0.2350X_{pn}X_{pe} + 0.5202X_{pn}X_{en} + 0.2225X_{pn}X_{an} - 0.3609X_{pe}X_{en} \\ &- 0.3518X_{pe}X_{an} - 0.0663X_{pe}X_{ab} - 0.3314X_{en}X_{an} + 0.1260X_{en}X_{ab} \\ &- 0.0334X_{an}X_{ab} \end{aligned} \tag{11}$$

4.1.1. Optimal factor setting and variability in energy system cost

In Fig. 6a, the electricity and natural gas price with the most important main effects are shown as two-factor relationships. The approximated energy system cost minimum, which is given annually in million EUR (MEUR/a), occurs with the optimal factor setting. The 3D surface plot gives a good overview of the variability in energy system cost when economic factor levels increase or decrease, and it shows how a factor's price change and its effect on energy system cost depends on the other factors' price level. The importance of the electricity and

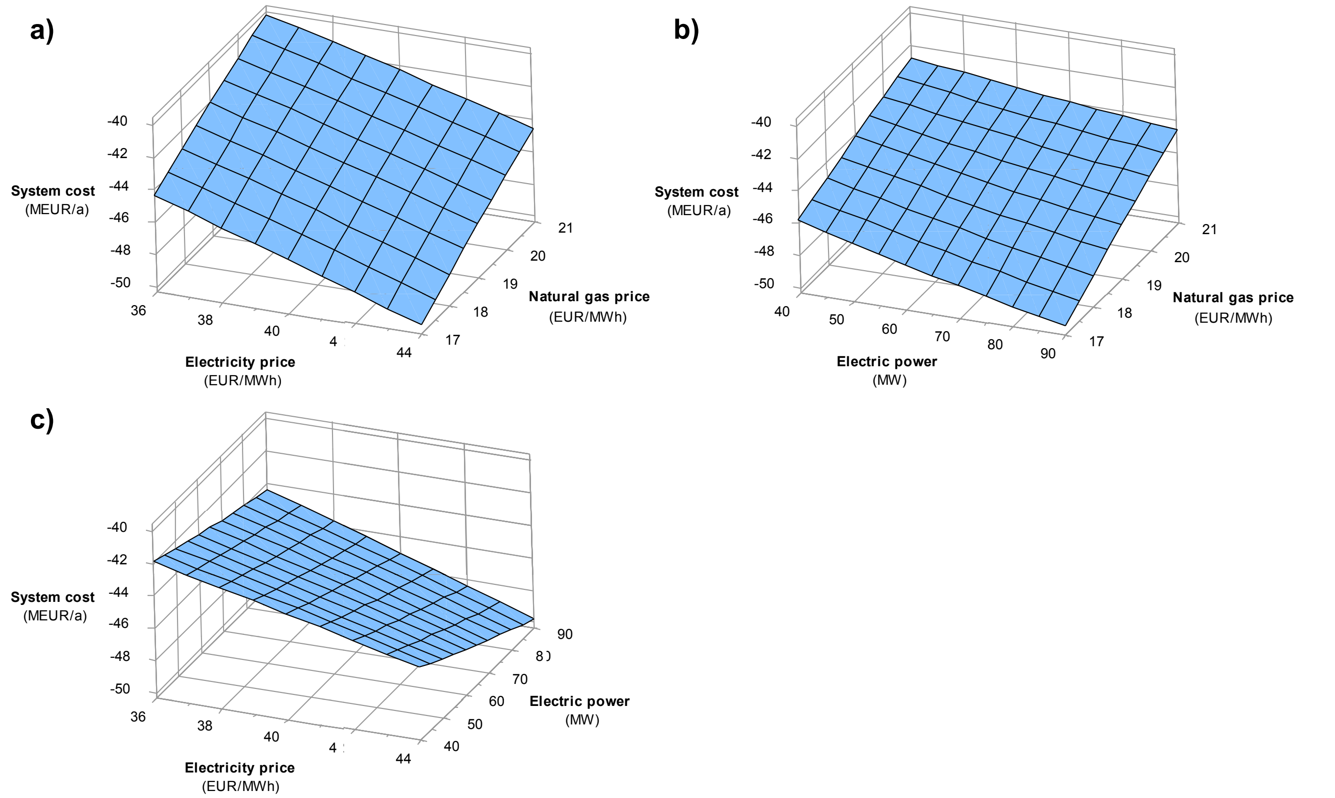


Fig. 6. 3D surface plots for response variable energy system cost with factor relationships natural gas price versus electricity price (a), and respective natural gas price (b) and electricity price (c) versus electric power NGCC CHP plant. Design factors with hold value levels vary for each plot and are price of biofuel (18 EUR/MWh), natural gas (17 EUR/MWh), electricity (44 EUR/MWh), and electric power size (90 MW) and alpha values (1.2, 0.52).

natural gas prices is explained by model assumptions and the cost optimization in MODEST. A decrease in fuel costs and increase in the electricity price makes a CHP plant operation more profitable. The factors' main and interaction effects contribute most to the reduction of energy system cost. The less important biofuel price may be explained by the relatively low output capacities of the biofuel CHP plants. The main effect of the biofuel price is lower than the largest interaction effect ($\beta_{pn,en}$). According to hold factor levels, a biofuel price change can be expected as linear and proportional to the level change through relatively low main and interaction effects and the lack of a squared effect, while changes in technical factors (X_{en}, X_{an}) with relevant squared and interaction effects may have some effect on the characteristics of the 3D surface plot.

4.1.2. CHP plants design characteristics and operation strategy

The metamodel designates a large electric power size and high alpha value as optimal solution for the NGCC CHP plant. The higher alpha value provides an operation strategy to maximize electricity output. The optimal solution is also strengthened by a negative interaction effect ($\beta_{en,an}$). The main and interaction effects from the electric power size are more substantial through its higher relative importance than effects from a higher alpha value ($\beta_{an}, \beta_{pn,an}, \beta_{pe,an}$). According to biofuel CHP plants, the improved electricity output capacity has only a minor effect on energy system cost, since the alpha value has the lowest main effect in the metamodel.

The designated output capacity of the NGCC CHP plant is robust for all energy price variations. As shown in Fig. 6b and c, the electric power size with 90 MW produces lowest energy system cost independent of natural gas and electricity prices. According to alpha values, the two-factor relationships $X_{pn}X_{an}$ and $X_{pe}X_{an}$ have similar surfaces as illustrated in Fig. 6b and c and show comparable behavior by changed hold factor levels. The reason for low alpha value (X_{ab}) effects can be expected to be like the biofuel price.

Changes in energy system cost are not only linear through the presence of largest squared and interaction effects. As shown in Fig. 7d–g, by changing the electricity and natural gas price to lowest and highest level, respectively, surface plots show less variability and more curvature with increased energy system cost. Furthermore, the plots indicate that the alpha value (X_{an}) has a low impact on energy system cost by an electric power size of 40 MW for the NGCC CHP plant. The operation strategy to maximize the electricity production is no longer preferred. Moreover, at the alpha value of 0.8, a larger electric power size seems to be less preferable due to increasing energy system cost, in particular with a low biofuel price and high alpha value for biofuel CHP plants (Fig. 7d). This effect is following the trade-offs in the metamodel.

4.1.3. Trade-offs in the metamodel

There are also trade-offs in the metamodel between using a large electric power size, low biofuel price, and high alpha value for biofuel CHP plants as shown by interactions $X_{en}X_{ab}$ and $X_{pb}X_{en}$. The optimal factor setting leads here to an increase in energy system cost. This needs further explanation. Studying the detailed scenario results of the 49 optimizations shows that the biofuel CHP plants' heat and electricity production increases substantially by an electric power size of 40 MW for the NGCC CHP plant. Together with a level combination of low biofuel price (X_{pb}) and high alpha value (X_{ab}), a further increase of the CHP plants' output is then provided. However, the interaction effects are relatively small compared with important economic and technical factor effects.

4.1.4. Validation of fitted metamodel with respect to MODEST model

The validation of the metamodel was performed with respect to the MODEST model by using additional scenarios (V1-V10) with new factor

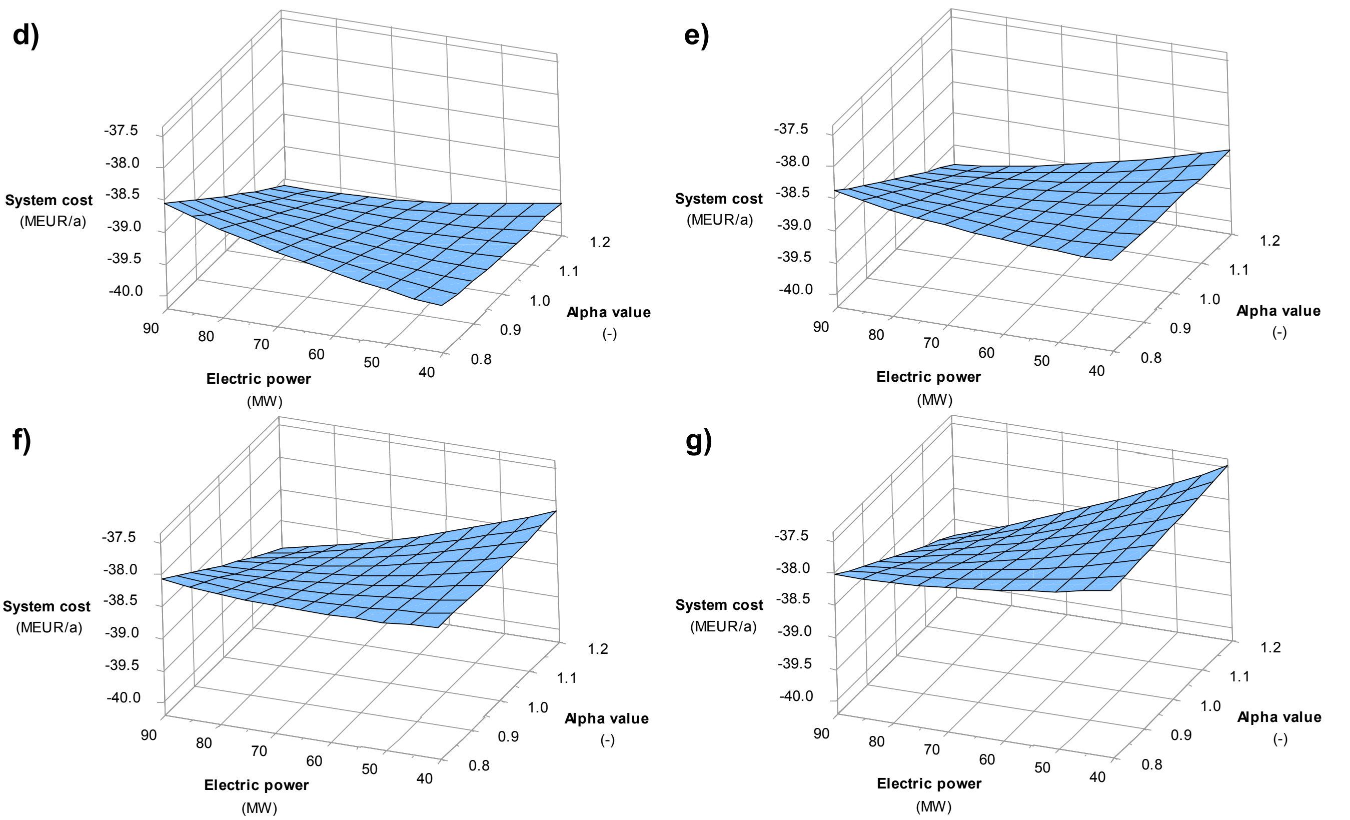


Fig. 7. 3D surface plots for response variable energy system cost with factor relationship electric power versus alpha value NGCC CHP plant. Factor level of alpha value for biofuel CHP plants changes from 0.52 (d, f) to 0.28 (e, g) and the level of biofuel price is 18 EUR/MWh in (d, e) and changes to 22 EUR/MWh in (f, g). Design factor levels for price of natural gas (21 EUR/MWh) and electricity (36 EUR/MWh) are hold constant. Note that the scale of system cost differs from previous plots.

settings (Table A4). Factor settings were randomly chosen within the design region, with exception of the first scenario (V1) including optimal level combinations.

In Fig. 8, the energy system costs predicted by the metamodel and optimized by the MODEST model are compared. Outputs are arranged in ascending order according to MODEST. The metamodel tends to underestimate lowest optimized energy system costs. A maximum deviation occurs in V1 with an ARE of 0.73%. In contrast, the metamodel

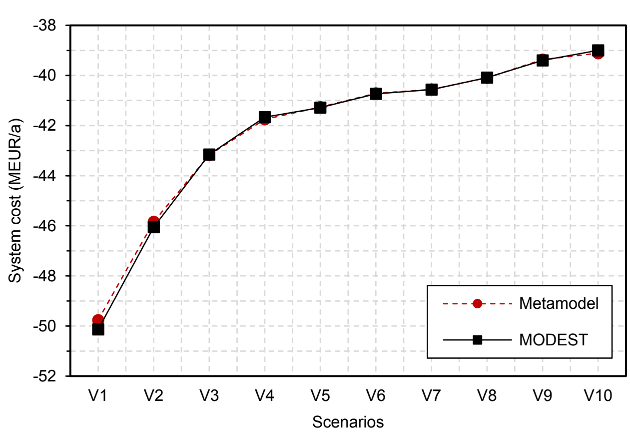


Fig. 8. Metamodel predictions and MODEST model outputs for annual energy system cost performed with additional scenarios (V1-V10).

seems to overestimate the highest optimized energy system cost in V10 resulting in an ARE of 0.29%. Similar systematic over- and underestimations of MODEST outputs can be seen in the fitting data set (run 1–49), but with relatively low deviations and a maximal ARE of 0.4%.

The underestimation in V1 may be explained by the level combination where all factors are at the same time either at their lowest or highest levels. Such a combination is basically avoided by the experimental design. Detailed result analysis of all scenarios shows that a remarkable increase in the electricity production with additional gained revenues occurs by a combination of X_{en} (1), X_{an} (1), X_{pe} (1), and X_{pn} (-1). Overestimations are less easy to explain. However, without an explicit maximum threshold for ARE, and with the consideration that model predictions are not the primary goal in the study, the metamodel is valid. Results for fitting and validation data set are found in Tables A2 and A4.

4.2. Comparison against system performance of reference and base case

Results for the comparison of system performances are generated by additional model runs with MODEST. For the regional PES case, the considered optimal factor setting is based on estimated main effects in the metamodel (Section 4.1). Reference and base case include corresponding biofuel and electricity price levels, which do not imply optimal price conditions for the respective case. Reference case contains accumulated results from separate systems in Gävle and Sandviken in subsections energy system cost and CO_2 emissions.

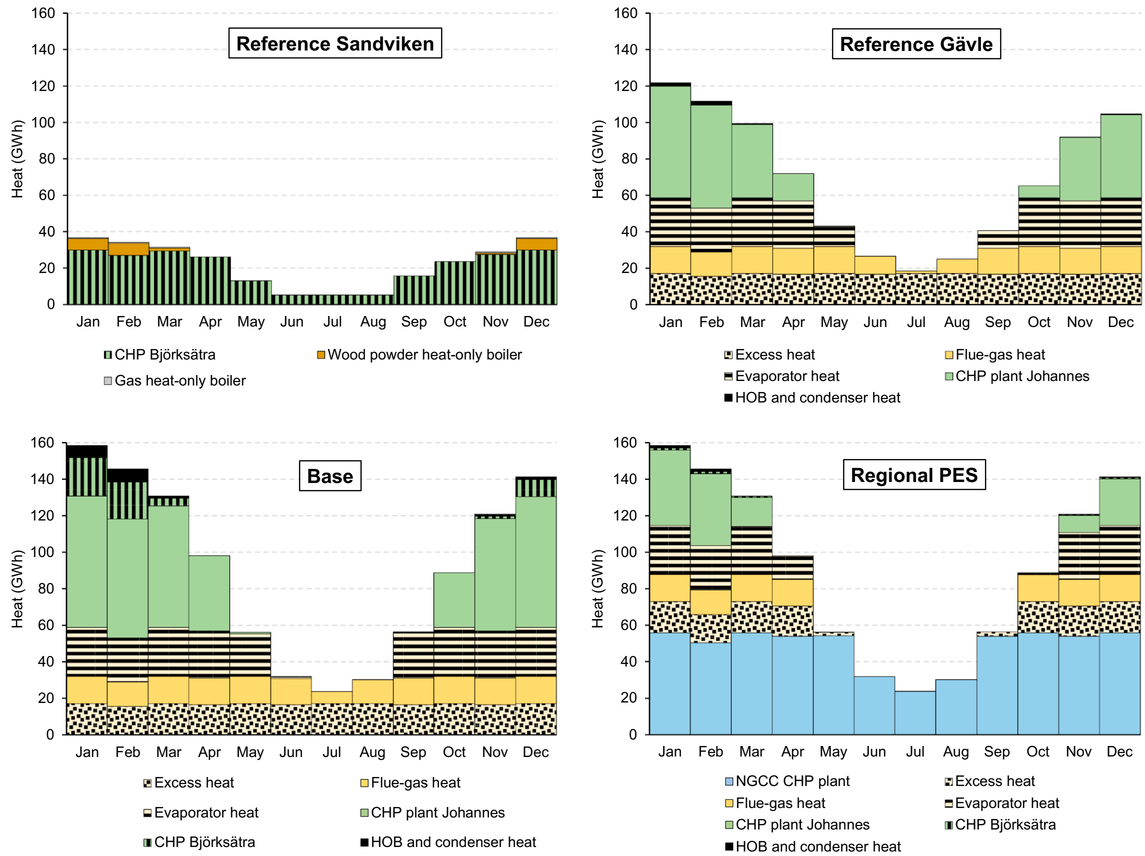


Fig. 9. Monthly-based heat production curves with industrial heat use for defined cases. Reference case represents separate district heating systems in Sandviken and Gävle. Base and regional PES case consider the district heating systems with existing energy supply as an interconnected regional system. The potential energy supply (PES) in regional PES case includes improved (Johannes, Björksätra) and new (NGCC) CHP plants. HOB and condenser heat for regional systems contain gas and wood powder heat-only boiler heat production from Sandviken.

4.2.1. Production of heat and electricity and the use of industrial heat

The CHP and HOB heat production and the use of supplied industrial heat are shown in Fig. 9. Monthly DH demand are aggregated numbers from modeled time steps and include distribution losses. The total use of heat supply sources in regional cases amounts here to 1082 GWh annually (reference case see Table 3). Illustrations reflect optimal heat supply with arranged merit order according to MODEST outputs, where sources with lowest costs are selected first to cover base heat loads and increased DH demands are then covered by sources with next highest cost and so on. Note that annual average prices used do not consider seasonal energy price variations, for instance, lower electricity prices during summer months through less demand which may affect CHP plant operations. HOB and condenser heat are aggregated outputs through relatively low amounts of produced and supplied heat. Besides the recovered FGC heat from biofuel CHP plants, the Björksätra CHP also includes heat-only production.

In Fig. 9, separate results for DH in Sandviken and Gävle (reference case) correspond to outputs shown in Table 3. Only the CHP electricity production in Sandviken increases here by 13 GWh/a through the higher electricity price. The reduced heat-only production is covered by the wood powder HOB (here 22 GWh/a). A system integration with merged heat loads (base case) provides an increased use of cost-effective heat supply sources. Heat supply sources covering base heat loads have increased utilization between May and September, for instance, low-priced FGC heat increases to 165 GWh/a. The increased utilization of the Johannes CHP plant between October and April is explained by the higher plant efficiency and larger FGC heat output capacity than the Björksätra CHP, which leads to less costly biofuel use, more recovered heat use, additional full-load operations (2300 h) from a relatively low level, and a heat and electricity production of 409 and 89 GWh/a, respectively. CHP Björksätra outputs are at the same time reduced to 56 GWh heat and 10 GWh electricity annually.

Results for the regional PES case in Fig. 9 show that the optimal factor setting leads to an NGCC CHP plant operation that dominates the heat supply for DH. The heat production amounts to 576 GWh/a with a full-load operation of about 5800 h. Only DH demand reductions in summer limit heat production. The operation strategy to produce more electricity results in an electricity output of 691 GWh/a. In contrast, the effect of improved biofuel CHP plants together with low biofuel and high electricity prices is limited, while not competitive to low-priced industrial heat supply. Only the Johannes CHP plant has a fully utilized electricity output capacity, but entirely on peak days with highest DH demand. It is notable that the CHP plant's annual heat production (131 GWh/a) is roughly half that in the reference case (Gävle), but the electricity output decreases only by 5 GWh to 51 GWh/a through the higher alpha value of 0.52. The increased electricity output capacity of

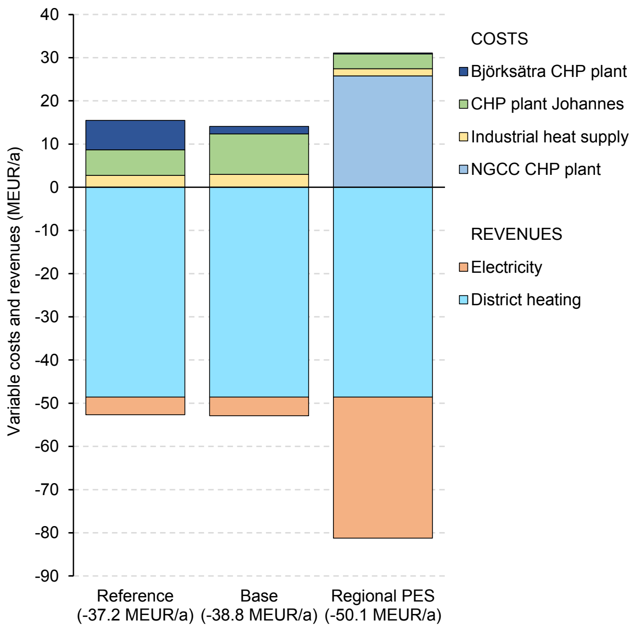


Fig. 10. Sum of annual variable costs for CHP plants' heat and electricity production and industrial heat supply and revenues for sales of electricity to the market and heat for district heating. Numbers in brackets are annual energy system cost with summarized costs and revenues for each defined case.

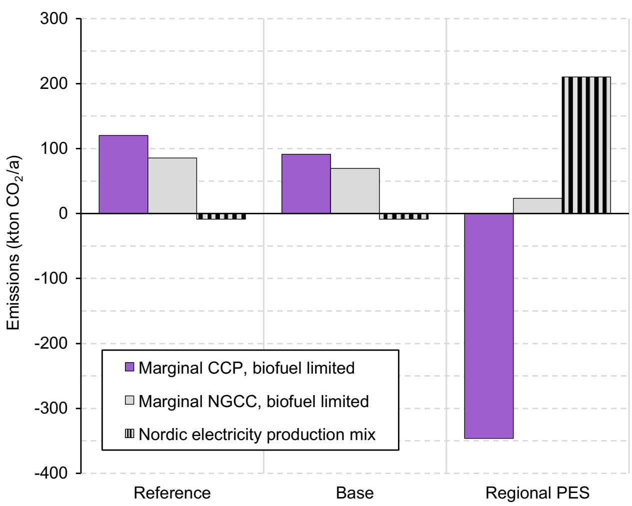


Fig. 11. Defined cases with total annual CO_2 emissions (kiloton per annum) accounted for direct and indirect emissions with marginal source of electricity (biofuel as limited resource and allocated with CO_2 emissions) and Nordic electricity production mix (biofuel as CO_2 neutral). Marginal sources are coal-condensing power (CCP) and NGCC-condensing power (NGCC) plants.

the Björksätra CHP is temporarily utilized on peak days, but almost negligible. Industrial heat supplies are here reduced to 121 GWh/a excess heat, 144 GWh/a evaporator heat, and 102 GWh/a FGC heat (cf. Table 3).

4.2.2. Energy system cost

For each case, the energy system cost is presented as annual variable costs and revenues in Fig. 10. Revenues from sale of electricity and heat for DH are generally higher than variable costs for CHP plants' heat and electricity production and the use of industrial heat. Energy system costs shown at the bottom of Fig. 10 therefore result in negative values entirely. Heat sales are constant for each case through same average heat price and DH demand. Electricity sales increase slightly from 4.1 to 4.3 MEUR/a through the system integration (base case) and are largest with 32.7 MEUR/a with the NGCC CHP plant operation (regional PES case). Variable costs decrease with regional DH through less costly CHP heat and electricity production, which together with an increased use of low-priced industrial heat are the main reason for the reduced energy system cost. The higher variable costs in the regional PES case occur mainly through the purchase of natural gas (77%) for the NGCC CHP plant's heat and electricity production.

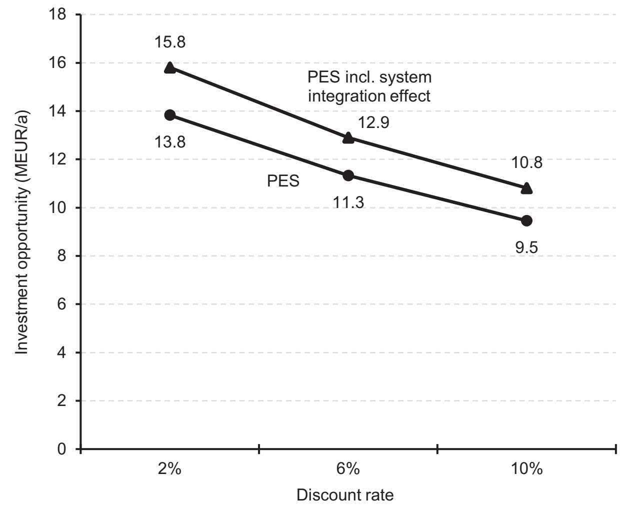


Fig. 12. Investment opportunity for PES (potential energy supply) with and without system integration effect based on different discount rates.

4.2.3. CO_2 emissions

The annual CO_2 emissions are calculated based on Table 4 and presented in Fig. 11 as a total of direct and indirect emissions. Results show considerable variation between cases with existing energy supply and PES, and among marginal and average accounting methods. According to reference and base case, CO_2 emissions are significantly higher for marginal electricity sources (CCP, NGCC) with biofuel as limited resource than for a Nordic electricity production mix considering biofuel use as CO_2 neutral. A system integration (base case) reduces CO_2 emissions for marginal sources through the decreased use of biofuel in the Björksätra CHP plant Sandviken, while a Nordic electricity production mix has almost unchanged negative CO_2 emissions (-9 kton CO_2/a), since calculated savings from the electricity production are larger than CO_2 emission from the use of LPG. According to the regional PES case, variations in CO_2 emissions occur mainly by the opportunity to credit CO_2 emission savings from the NGCC CHP plant's electricity production. Only a higher emission factor for marginal CCP accounting leads to negative emissions. Emissions from the use of natural gas and the use of biofuel in the CHP plants is larger than savings from the electricity production in marginal NGCC. The Nordic electricity production mix with biofuel use as CO_2 neutral has the highest CO_2 emissions here through the use of natural gas. Only marginal electricity sources show adequate savings to compensate for increased direct CO_2 emissions from natural gas combustion. Marginal emission savings have generally smaller impact on a long-term (NGCC) than on a short-term (CCP) perspective. The difference depends on lower CO_2 emission factors allocated for the biofuel release and a more efficient electricity production of NGCC-condensing power plants. In comparison with the reference case, largest marginal CO_2 emission savings occur with 466 kton/a (CCP, biofuel limited) and 62 kton/a (NGCC, biofuel limited) in the regional PES case, but also highest CO_2 emissions with 210 kton/a and a Nordic electricity production mix.

4.3. Investment opportunity

The investment opportunity for improved biofuel CHP plants and new NGCC CHP plant is presented in Fig. 12 as PES with and without system integration effect. A system integration effect here is the difference between lower/upper investment opportunities for respective discount rate. The investment opportunity with a 6% discount rate corresponds to energy system cost differences in Fig. 10. The results show that investment opportunities vary substantially if lower (2%) and higher (10%) investment risks are associated with PES, for instance, by considering that a high-risk investment may command a discount rate of 10% and reduce the investment opportunity to 9.5 MEUR/a. Investment opportunities rise by the inclusion of system integration effects, but this also requires investments in system interconnection. Variations in the system integration effect are generally lower and range between 1.3 (10%) and 2 MEUR/a (2%).

However, by knowing the variability in energy system cost as illustrated previously in Fig. 6a, uncertainties associated with an investment in a capital-intensive new plant are reasonable. With large amounts of purchased natural gas and a relatively high share of gained revenues from electricity sales (cf. Fig. 10), a rising natural gas price and falling electricity price may considerably increase energy system costs and consequently decrease the profitability of investing in a NGCC CHP plant, particularly at higher discount rates. In addition, by considering fixed costs together with Swedish operation taxes for natural gas use and emission rights, investment in a new plant may not be profitable. According to improved biofuel CHP plants, the impact of changing energy prices is comparatively low through a minor contribution of heat and electricity sales on revenues. But incentives to invest in the Björksätra CHP plant with low utilization grade may not exist (cf. Fig. 9), while an investment in the Johannes CHP plant may be advantageous if less heat and more electricity outputs are of interest.

5. Further considerations to develop the energy system study

The consideration of varying DH demands, different heat sale prices, biogas utilization and their impact on energy system cost and design may be of interest for further studies. Considering DH demand variations is of importance for analyzing uncertainties in changing heat loads. A typical annual DH demand reflects only average heat loads usable to dimension heat production plants, but real heat loads vary from year to year depending on outdoor temperatures. Such variations lead to deviations from average values and less stable conditions for investment opportunities. Besides the price of electricity, heat sale price is an important market value for cost-effective CHP production. A change of the assumed average heat price should be considered through the new mix of heat supply sources with different variable costs and the regional DH with merged heat loads and changed distribution costs. Replacing natural gas with renewable biogas is an important aspect for achieving sustainable energy systems and reducing direct CO_2 emissions. A cost-effective biogas utilization may depend on its price, which in turn will depend on scale of production and incentives including different types of policies.

6. Conclusions

The influence of the price of biofuel, natural gas, and electricity on CHP production with improved and new CHP plants in a DH system was investigated by using an optimization tool and metamodel.

The results show that a polynomial regression metamodel requires only a relatively small number of scenarios to identify the significance of techno-economic factor effects. A factor's significance can be directly identified from coefficients in the regression metamodel (Eq. (11)) or illustrated by graphical solutions (Figs. 6 and 7). A standardizing of factor levels provides a simultaneous study of technical and economic input variables on a common scale. To study the entire design region where factors take values gives a more thorough understanding than using single factor value levels. Applications with Design of Experiment and response surface models are relatively simple through the access to statistical software. The results also show that main effects of electricity and fuel prices are important, but that variations in energy system cost can also be caused by many cross-factor interactions with technical factors.

The main limitation for the combination methods used is that adequate input data are necessary for reliable metamodel approximations, since a wider factor value range can lead to decreased approximation quality, especially by significant variations in the energy supplies caused by the model optimization to determine optimal system design and operation. The use of many factors may lead to comprehensive energy system studies, and an initial factor screening may help here.

According to system performance analysis with defined cases and optimal design factor setting and based on given techno-economic assumptions, the following main conclusions can be drawn:

- Utilizing a NGCC CHP plant with large electric power size and high alpha value significantly increases the electricity production, here by up to 650 GWh annually, but the use of industrial heat decreases substantially at the same time.
- New plant installation depends highly on investment risk and electricity and fuel market prices, whereas a high-risk investment and increased energy system cost markedly decrease the profitability.
- Improving the alpha value of biofuel CHP plants may be seen as an option to increase the electricity production in DH systems.
- CO_2 emission savings can be reached by 466 kton annually if marginal electricity production from coal-condensing power plants is avoided and biofuel is released at the same time, but emissions will be substantially higher if allocated with a Nordic electricity system perspective or more efficient marginal electricity production.
- According to heat loads, connecting DH systems provides the increased use of recovered excess heat from CHP plants or industrial processes.

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Appendix A

See Tables A1–A4.

Table A1
Annual time division with system heat loads (MW) for district heating in Gävle and Sandviken. Merged numbers represent heat loads and demand for regional district heating.

Season (month)[a]	Type of day	Diurnal period	Hours	Heat load[b]		
				Sandviken	Gävle	Merged[c]
Nov-Mar	Weekday	1	06-07	42	136	178
		2	07-08	42	137	179
		3	08-16	40	125	164
		4	16-22	41	131	171
		5	22-06	38	125	163
	Weekend	6	06-22	39	126	165
		7	22-06	41	125	165
	Peak day	8	06-07	69	184	254
		9	07-08	91	188	279
		10	08-16	87	189	276
		11	16-22	87	200	287
		12	22-06	57	185	242
Apr, Sept, Oct	Weekday	1	06-22	26	71	97
		2	22-06	27	75	102
	Weekend	3	06-22	26	67	93
		4	22-06	26	75	100
May, Jun, Aug	Weekday	1	06-22	10	36	46
		2	22-06	10	40	49
	Weekend	3	06-22	8	38	46
		4	22-06	9	41	50
Jul	Weekday	1	06-22	6	22	28
		2	22-06	7	22	29
	Weekend	3	06-22	6	21	27
		4	22-06	6	21	27
	District heating demand (GWh/a)			231	721	952

[a] Number of days and hours can vary for each month.
[b] Average values for respective season and diurnal period.
[c] Maximal heat load occur on a peak day in January with 334 MW.

Table A2
Energy system cost (MEUR/a) for each performed scenario (run 1–49) generated by the MODEST model and predicted by the metamodel. ARE is the calculated Absolute Relative Error (%).

Run	MODEST	Metamodel	ARE	**Run**	MODEST	Metamodel	ARE
1	−42.3852	−42.4273	0.10	**26**	−39.2771	−39.2153	0.16
2	−40.9982	−41.0571	0.14	**27**	−41.5820	−41.5831	0.00
3	−40.2242	−40.2159	0.02	**28**	−41.1716	−41.1666	0.01
4	−38.8373	−38.8457	0.02	**29**	−41.7801	−41.7890	0.02
5	−45.2379	−45.2334	0.01	**30**	−40.3118	−40.3136	0.00
6	−44.7004	−44.7117	0.03	**31**	−44.1528	−44.2175	0.15
7	−40.9962	−40.9412	0.13	**32**	−43.4976	−43.5906	0.21
8	−40.4587	−40.4195	0.10	**33**	−42.0456	−42.0568	0.03
9	−40.8551	−40.7895	0.16	**34**	−39.2435	−39.2500	0.02
10	−38.2984	−38.4526	0.40	**35**	−44.3597	−44.3015	0.13
11	−43.8555	−43.9179	0.14	**36**	−40.5831	−40.6047	0.05
12	−40.6882	−40.6412	0.12	**37**	−42.5374	−42.5564	0.04
13	−42.3096	−42.3972	0.21	**38**	−39.7353	−39.7496	0.04
14	−39.2768	−39.1703	0.27	**39**	−44.9850	−44.9345	0.11
15	−47.0465	−46.9329	0.24	**40**	−41.2083	−41.2377	0.07
16	−42.7446	−42.7662	0.05	**41**	−40.3061	−40.3219	0.04
17	−38.9421	−38.9157	0.07	**42**	−39.4634	−39.4391	0.06
18	−41.3650	−41.4236	0.14	**43**	−43.5367	−43.5514	0.03
19	−40.6281	−40.6359	0.02	**44**	−42.6675	−42.6687	0.00
20	−44.6156	−44.5872	0.06	**45**	−40.8049	−40.8188	0.03
21	−39.5796	−39.6015	0.06	**46**	−39.8230	−39.8096	0.03
22	−42.3895	−42.3745	0.04	**47**	−44.2872	−44.3135	0.06
23	−40.8827	−40.8177	0.16	**48**	−43.3045	−43.3044	0.00
24	−45.0149	−45.0341	0.04	**49**	−41.6968	−41.6764	0.05
25	-40.5695	-40.4803	0.22				

Table A3
Analysis of variance for the regression metamodel developed for energy system cost.

Source	Degrees of freedom (df)	Sum of squares (SS)	Mean squares (MS)	*F*-Value	*P*-Value
Model	23	196.100	8.5261	1743.45	0.000
Linear	6	188.238	31.3730	6415.28	0.000
X_{pb}	1	5.369	5.3692	1097.92	0.000
X_{pn}	1	63.445	63.4452	12973.54	0.000
X_{pe}	1	67.824	67.8243	13869.01	0.000
X_{en}	1	28.775	28.7747	5883.97	0.000
X_{an}	1	20.900	20.9002	4273.76	0.000
X_{ab}	1	1.924	1.9243	393.50	0.000
Square	5	0.596	0.1193	24.39	0.000
$X_{pn}*X_{pn}$	1	0.196	0.1962	40.12	0.000
$X_{pe}*X_{pe}$	1	0.043	0.0428	8.74	0.007
$X_{en}*X_{en}$	1	0.065	0.0653	13.35	0.001
$X_{an}*X_{an}$	1	0.006	0.0057	1.16	0.291
$X_{ab}*X_{ab}$	1	0.006	0.0058	1.19	0.286
2-Way Interaction	12	7.266	0.6055	123.81	0.000
$X_{pb}*X_{en}$	1	0.720	0.7200	147.22	0.000
$X_{pb}*X_{an}$	1	0.022	0.0221	4.52	0.043
$X_{pb}*X_{ab}$	1	0.008	0.0080	1.63	0.213
$X_{pn}*X_{pe}$	1	0.442	0.4417	90.31	0.000
$X_{pn}*X_{en}$	1	2.165	2.1649	442.68	0.000
$X_{pn}*X_{an}$	1	0.792	0.7920	161.96	0.000
$X_{pe}*X_{en}$	1	1.042	1.0418	213.03	0.000
$X_{pe}*X_{an}$	1	0.990	0.9902	202.49	0.000
$X_{pe}*X_{ab}$	1	0.070	0.0703	14.38	0.001
$X_{en}*X_{an}$	1	0.879	0.8788	179.69	0.000
$X_{en}*X_{ab}$	1	0.127	0.1270	25.97	0.000
$X_{an}*X_{ab}$	1	0.009	0.0089	1.82	0.189
Error	25	0.122	0.0049		
Total	48	196.222			

Table A4
Validation data set for scenarios (run) with new design factor settings (V1-V10) and results for energy system cost (MEUR/a) generated by the MODEST model and predicted by the metamodel. ARE is the calculated Absolute Relative Error (%).

Run	x_{pb}	x_{pn}	x_{pe}	x_{en}	x_{an}	x_{ab}	MODEST	Metamodel	ARE
V1	18	17	44	90	1.2	0.52	−50.1325	−49.7651	0.73
V2	20	18	42	80	1.2	0.45	−46.0562	−45.8418	0.47
V3	19	20	42	80	1.1	0.45	−43.1516	−43.1715	0.05
V4	19	17	40	40	1.0	0.30	−41.6653	−41.7497	0.20
V5	21	20	41	70	1.0	0.40	−41.2799	−41.2645	0.04
V6	22	18	37	60	1.1	0.35	−40.7335	−40.7131	0.05
V7	18	19	38	50	0.9	0.40	−40.5655	−40.5560	0.02
V8	20	21	40	60	1.0	0.40	−40.0837	−40.0911	0.02
V9	20	19	36	50	1.0	0.30	−39.3969	−39.3671	0.08
V10	19	21	39	40	0.8	0.35	−38.9969	−39.1104	0.29

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